Universal Scale Locking and the Turok-36 Test

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral; see RECOVERY_PROVENANCE.md.

16. A conventional hidden-sector generator

A safe infrared proof of principle uses

$$G = G_{\rm SM} \times SU(2)_X,$$

with the Standard Model Higgs doublet H and a hidden scalar doublet Φ:

$$h = \sqrt{2H^\dagger H}, \qquad \chi = \sqrt{2\Phi^\dagger \Phi}.$$

The Jordan-frame action includes

$$\frac{1}{2}(\xi_\chi \chi^2 + \xi_H h^2)R,$$

ordinary two-derivative kinetic terms, the hidden gauge field, and

$$V = \frac{\lambda_\chi}{2}(\Phi^\dagger\Phi)^2 - \lambda_p (H^\dagger H)(\Phi^\dagger \Phi) + \frac{\lambda_H}{2}(H^\dagger H)^2.$$

Hidden gauge loops generate a Coleman-Weinberg scale $f = \langle \chi \rangle$. The portal valley gives

$$h^2 = \frac{\lambda_p}{\lambda_H}\chi^2 \equiv \zeta_h^2 \chi^2,$$

and the gravitational coefficient becomes

$$F = \xi_{\rm eff}\chi^2, \qquad M_{\rm P}^2 = \xi_{\rm eff} f^2.$$

This construction is not proposed as the final UV theory. Its role is to demonstrate that a conventional, unitary, two-derivative matter sector can generate and transmit one common scale [@HiddenSU22013; @CurvedSM2018].

Exact common mode and fifth-force cancellation

In the Einstein frame,

$$m_A^E = \frac{M_{\rm P}^{(0)} m_A(\chi)}{\sqrt{F(\chi)}}.$$

If $m_A = c_A \chi$ and $F = \xi_{\rm eff}\chi^2$, then

$$m_A^E = \frac{c_A M_\mathrm{P}^{(0)}}{\sqrt{\xi_\mathrm{eff}}},$$

independent of the radial field. Therefore

$$\alpha_A \equiv \partial_\varphi \ln m_A^E = 0.$$

Exact universal scaling produces no composition-dependent fifth force. Imperfect locking gives

$$\alpha_A - \alpha_B = \partial_\varphi \ln \frac{m_A}{m_B} = \partial_\varphi \ln \frac{\omega_A}{\omega_B}.$$

Differential fifth-force charge and differential clock drift are the same derivative obstruction [@ScaleFifthForce2016; @ScaleFifthForce2021].

17. Turok's 36 fields as an optional microscopic origin

Boyle and Turok proposed 36 dimension-zero fourth-order scalar fields in a programme involving anomaly cancellation, vacuum energy, and an emergent Higgs [@BoyleTurok2022]. A possible chronometric connection is that the 36-field sector might generate one collective infrared singlet χ, for example through a mass gap or an invariant composite related to

$$\mathcal{O}_2 = \frac{1}{36} G_{IJ} g^{\mu\nu} \nabla_\mu \varphi^I \nabla_\nu \varphi^J.$$

The mapping would be

$$\text{36 microscopic dimension-zero fields} \longrightarrow \text{1 dimensionful infrared order parameter.}$$

However, an elementary propagator with leading behaviour $1/p^4$ cannot possess a nontrivial conventional positive Källén-Lehmann spectral measure. If

$$G(Q^2) = \int_0^\infty ds\, \frac{\rho(s)}{Q^2 + s}, \qquad \rho(s) \geq 0,$$

then the leading term is

$$G(Q^2) = \frac{1}{Q^2} \int \rho(s) ds + O(Q^{-4}).$$

To begin at $1/Q^4$ requires $\int \rho = 0$, which with positivity forces $\rho = 0$. This does not exclude an indefinite-metric formalism, a gauge redundancy, or a healthy composite pole. It means physical positivity must be demonstrated for invariant observables, not inferred from the elementary fourth-order field.

Bateman and Turok propose a Krein-space and hidden-ghost-parity resolution, while Cline and Hell argue that ghosts, unitarity violation, and a confining fifth force remain [@BatemanTurok2026; @ClineHell2026]. The chronometric programme therefore treats the 36-field theory as an optional UV candidate subject to four mandatory tests:

1. Lorentzian unitarity on a physical state space;
2. positive spectral density or an equally strong reconstruction theorem for an invariant composite;
3. a stable condensate with $Z_\chi > 0$ and real dispersion;
4. acceptable universal matter coupling without a macroscopic confining force.

The safe low-energy construction survives whether or not that candidate passes.